



Feedback form for the course

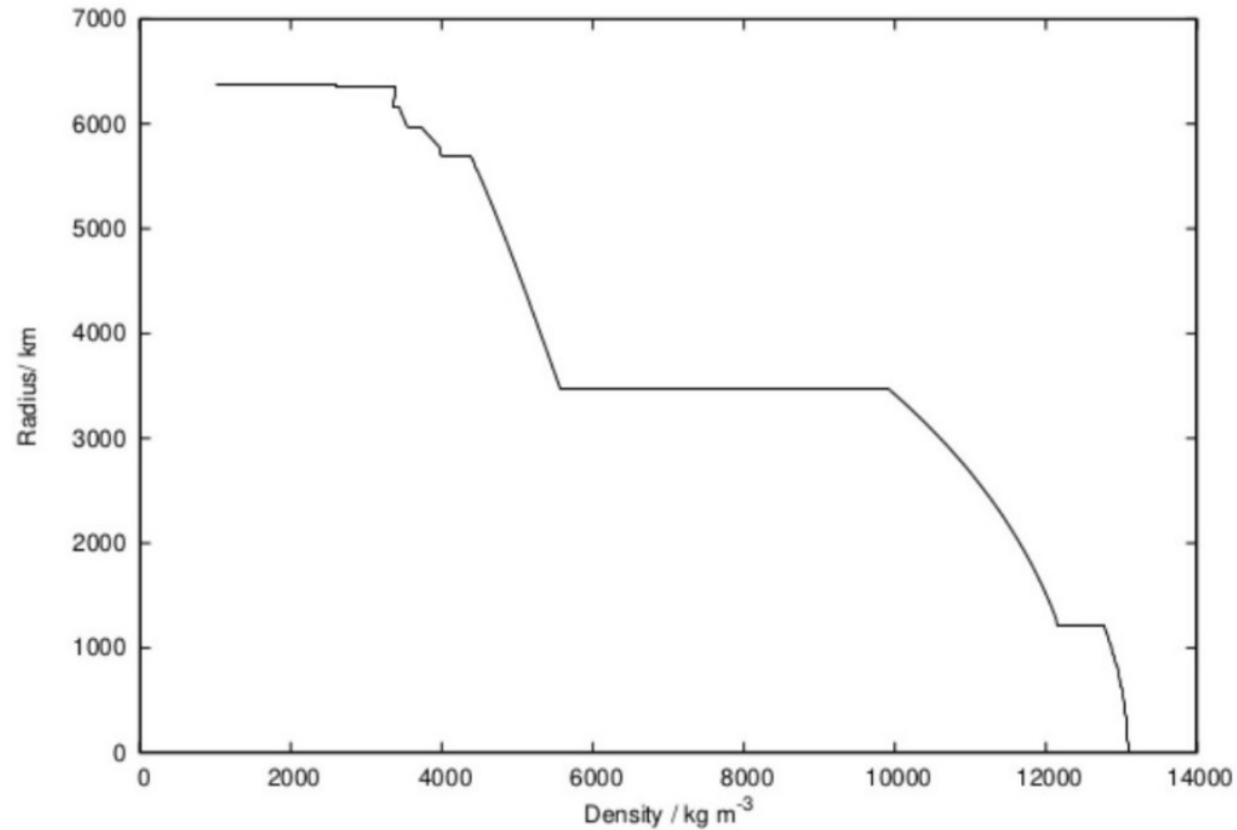
Please do fill this in before the end of term

We do take comments seriously, and they help to make the course better over time

https://cambridge.eu.qualtrics.com/jfe/form/SV_3z0oG7tq3O9xtpc

Free oscillations of the Earth

From last lecture



Dziewonski & Anderson (1981)

Operator formalism

Equations of motion

Linearised equations with self-gravitation and rotation:

$$\rho \left(\frac{\partial^2 u_i}{\partial t^2} + 2\epsilon_{ijk}\Omega_j \frac{\partial u_k}{\partial t} + \epsilon_{ijk}\epsilon_{klm}\Omega_j\Omega_l u_m \right) - \frac{\partial}{\partial x_j} \left(A_{ijkl} \frac{\partial u_k}{\partial x_l} \right) - \rho\gamma_i^1 = -\frac{\partial \bar{S}_{ij}}{\partial x_j},$$

$$\gamma_i^1 = -G \int_M \rho' \left(\frac{\delta_{ij}}{[(x_k - x'_k)(x_k - x'_k)]^{3/2}} - \frac{3(x_i - x'_i)(x_j - x'_j)}{[(x_k - x'_k)(x_k - x'_k)]^{5/2}} \right) (u_j - u'_j) d^3\mathbf{x}',$$

$$A_{ijkl} \frac{\partial u_k}{\partial x_l} \hat{n}_j = \bar{S}_{ij} \hat{n}_j$$

Simplifying notations will help a lot!

Weak form of the equations

For an arbitrary **test function**, we have:

$$\int_M \left[\rho \left(\frac{\partial^2 u_i}{\partial t^2} + 2\epsilon_{ijk}\Omega_j \frac{\partial u_k}{\partial t} + \epsilon_{ijk}\epsilon_{klm}\Omega_j\Omega_l u_m \right) - \frac{\partial}{\partial x_j} \left(A_{ijkl} \frac{\partial u_k}{\partial x_l} - \bar{S}_{ij} \right) - \rho\gamma_i^1 \right] w_i^* d^3\mathbf{x} = 0.$$

Integrating by parts and using the boundary conditions:

$$\int_M \left[\rho \left(\frac{\partial^2 u_i}{\partial t^2} + 2\epsilon_{ijk}\Omega_j \frac{\partial u_k}{\partial t} + \epsilon_{ijk}\epsilon_{klm}\Omega_j\Omega_l u_m - \gamma_i^1 \right) w_i^* + \left(A_{ijkl} \frac{\partial u_k}{\partial x_l} - \bar{S}_{ij} \right) \frac{\partial w_i^*}{\partial x_j} \right] d^3\mathbf{x} = 0.$$

If the latter identity holds for all test functions, then we can recover both the equations of motion and the boundary conditions

This is the **weak formulation** of the problem

Weak and strong formulations are equivalent up to technicalities

Kinetic energy form

We define the **kinetic energy form**:

$$\langle \mathbf{w} | P | \mathbf{u} \rangle = \int_M \rho w_i^* u_i \, d^3 \mathbf{x},$$

Acceleration term in the weak form can then be written:

$$\langle \mathbf{w} | P | \ddot{\mathbf{u}} \rangle.$$

This **sesqui-linear form** is associated with the linear operator:

$$(P\mathbf{u})_i = \rho u_i$$

Coriolis form

We define the **Coriolis form**:

$$\langle \mathbf{w} | W | \mathbf{u} \rangle = \int_M 2\rho \epsilon_{ijk} w_i^* \Omega_j u_k \, d^3\mathbf{x}$$

Associated with the linear operator:

$$(W\mathbf{u})_i = 2\rho \epsilon_{ijk} \Omega_j u_k$$

Potential energy form

We define the **potential energy form**:

$$\langle \mathbf{w} | H | \mathbf{u} \rangle = \int_M \left(\rho \epsilon_{ijk} \epsilon_{klm} w_i^* \Omega_j \Omega_l u_m + A_{ijkl} \frac{\partial w_i^*}{\partial x_j} \frac{\partial u_k}{\partial x_l} - \rho w_i^* \gamma_i^1 \right) d^3 \mathbf{x},$$

Along with the linear form linked to the stress glut:

$$\langle \mathbf{w} | \mathbf{f} \rangle = \int_M \bar{S}_{ij} \frac{\partial w_i^*}{\partial x_j} d^3 \mathbf{x}.$$

These two terms incorporate **both** the equations of motion and the boundary conditions

Equations of motion

The weak form of the equations of motion can now be written:

$$\langle \mathbf{w} | P | \ddot{\mathbf{u}} \rangle + \langle \mathbf{w} | W | \dot{\mathbf{u}} \rangle + \langle \mathbf{w} | H | \mathbf{u} \rangle = \langle \mathbf{w} | \mathbf{f} \rangle,$$

which is to hold for all test functions, $\mathbf{w}$

We can formally think of the problem as

$$P\ddot{\mathbf{u}} + W\dot{\mathbf{u}} + H\mathbf{u} = \mathbf{f}.$$

but the boundary conditions are then left implicit

Properties of the forms

The kinetic energy form is **self-adjoint**:

$$\langle \mathbf{w} | P | \mathbf{u} \rangle = \langle \mathbf{u} | P | \mathbf{w} \rangle^*,$$

The Coriolis form is **anti-self-adjoint**:

$$\langle \mathbf{w} | W | \mathbf{u} \rangle = \int_M 2\rho \epsilon_{ijk} w_i^* \Omega_j u_k \, d^3\mathbf{x} = - \left(\int_M 2\rho \epsilon_{ijk} u_i^* \Omega_j w_k \, d^3\mathbf{x} \right)^* = - \langle \mathbf{u} | W | \mathbf{w} \rangle^*,$$

Properties of the forms

The potential energy form is **self-adjoint**:

$$\langle \mathbf{w} | H | \mathbf{u} \rangle = \int_M \left(\rho \epsilon_{ijk} \epsilon_{klm} w_i^* \Omega_j \Omega_l u_m + A_{ijkl} \frac{\partial w_i^*}{\partial x_j} \frac{\partial u_k}{\partial x_l} - \rho w_i^* \gamma_i^1 \right) d^3 \mathbf{x},$$

Elastic term is clear due to hyperelastic symmetric

Centrifugal term follows by a simple calculation – details in handout

Gravitational potential term takes more work but proceeds in the same manner – details are not in handout and are not examinable

Non-rotating eigenvalue problem

Free oscillations

For long-period seismology, rotation is a relatively small term that is often handled through perturbation theory

Weak form in a non-rotating planet:

$$\langle \mathbf{w} | P | \ddot{\mathbf{u}} \rangle + \langle \mathbf{w} | H | \mathbf{u} \rangle = \langle \mathbf{w} | \mathbf{f} \rangle,$$

Free oscillations are time-harmonic solutions with no force term:

$$u_i(\mathbf{x}, t) = s_i(\mathbf{x}) e^{i\omega t}$$
$$-\omega^2 \langle \mathbf{w} | P | \mathbf{s} \rangle + \langle \mathbf{w} | H | \mathbf{s} \rangle = 0.$$

This is an eigenvalue problem, with squared eigenfrequency as the eigenvalue

Standard properties

You need to know how to show that:

- 1) Squared eigenfrequencies are real
- 2) Eigenfunctions in distinct eigenspaces are orthogonal:

$$\langle \mathbf{s}_1 | P | \mathbf{s}_2 \rangle = 0$$

Further properties to know but without proof:

- 1) Eigenspaces are all finite dimensional
- 2) Eigenfunctions form a complete basis for vector fields satisfying the traction-free boundary conditions

Eigenfunction expansions

We use k to label the eigenspaces, and m the eigenfunctions in each eigenspace

Within each eigenspace, we can choose an orthonormal basis, and hence:

$$\langle km|P|k'm'\rangle = \delta_{kk'}\delta_{mm'},$$

Completeness of the eigenfunctions means that we can write:

$$\mathbf{u} = \sum_{km} \langle km|P|\mathbf{u}\rangle |km\rangle,$$

This is the **eigenfunction expansion** of the vector field

Eigenfunction expansions

Equations of motion in the frequency-domain:

$$-\omega^2 \langle \mathbf{w} | P | \tilde{\mathbf{u}} \rangle + \langle \mathbf{w} | H | \tilde{\mathbf{u}} \rangle = \langle \mathbf{w} | \tilde{\mathbf{f}} \rangle.$$

Expanding the solution in the eigenfunction basis:

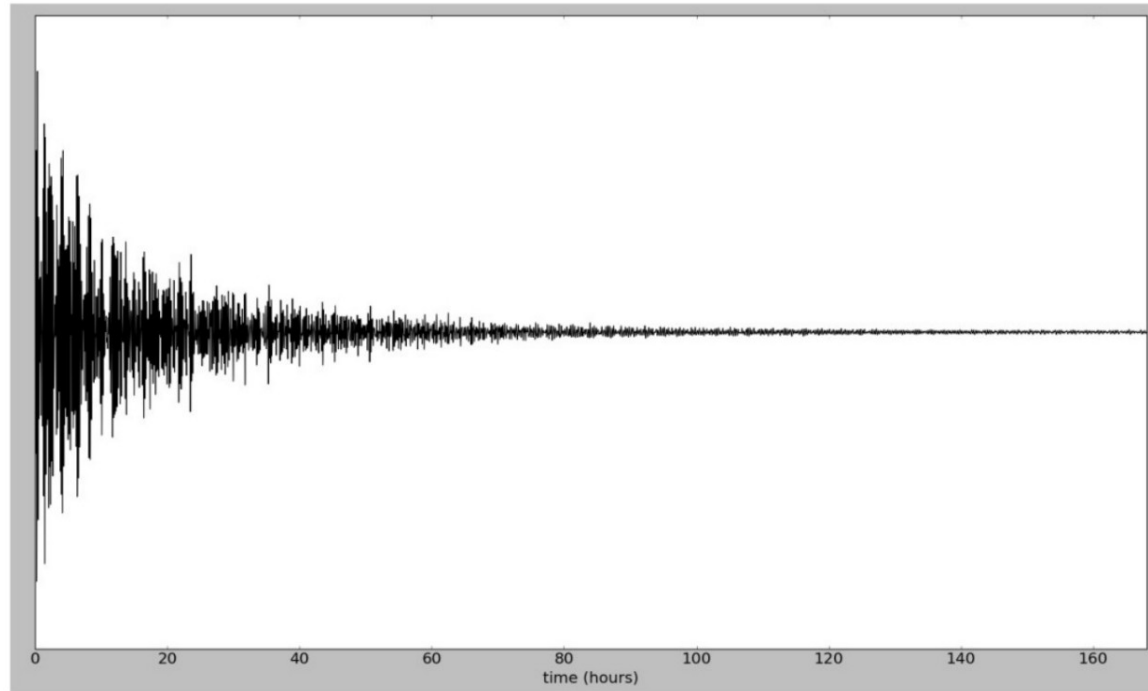
$$\tilde{\mathbf{u}} = \sum_{k'm'} \langle k'm' | P | \tilde{\mathbf{u}} \rangle |k'm'\rangle,$$

Frequency-domain solution readily derived:

$$\tilde{\mathbf{u}} = \sum_{km} \frac{\langle km | \tilde{\mathbf{f}} \rangle}{\omega_k^2 - \omega^2} |km\rangle,$$

Time-domain solution obtained by an inverse Fourier transformation – but details are not covered and not examinable

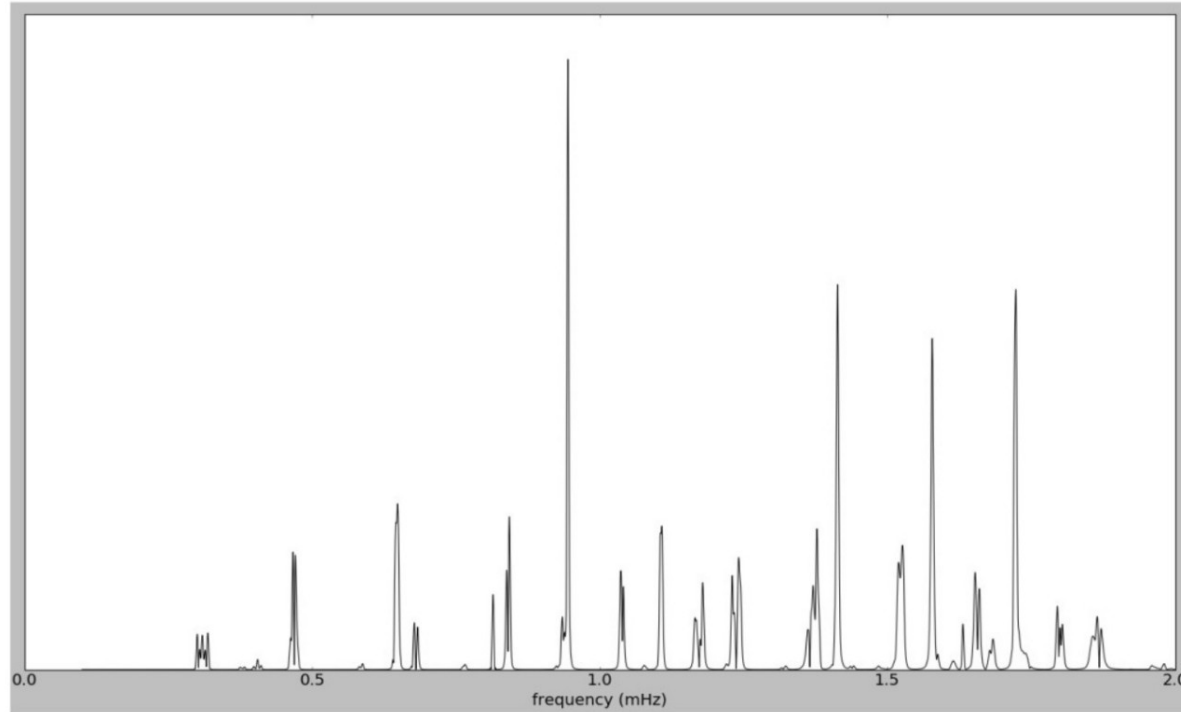
Long time-series



Several day long time-series following a large earthquake (Bolivia 1994)

The effects of attenuation are clearly seen within real data

Corresponding amplitude spectra



Resonance peaks (or poles) associated with each eigenfrequency

For good resolution in the frequency-domain, we need **long time series**

Gravitational stability

Rayleigh quotient

The non-rotating eigenvalue problem:

$$-\omega^2 \langle \mathbf{w} | P | \mathbf{s} \rangle + \langle \mathbf{w} | H | \mathbf{s} \rangle = 0.$$

Taking the test function to equal the eigenfunction:

$$\omega_k^2 = \langle km | H | km \rangle.$$

Sign of the squared eigenfrequency determined by the potential energy form

Positive values imply **stable** oscillations about equilibrium

Negative values imply **unstable**, exponentially growing perturbations

Zero values correspond to **linear growth in time** (double pole at the origin)

Non-gravitating case

Potential energy form simplifies such that:

$$\langle \mathbf{u} | H | \mathbf{u} \rangle = \frac{1}{4} \int_M A_{ijkl} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right)^* \left(\frac{\partial u_k}{\partial x_l} + \frac{\partial u_l}{\partial x_k} \right) d^3 \mathbf{x},$$

Vanishes for linearised rigid body motions:

$$u_i(\mathbf{x}) = a_i + A_{ij}x_j,$$

Six-dimensional eigenspace at zero-frequency, but not excited by earthquakes:

$$\langle \mathbf{w} | \mathbf{f} \rangle = \int_M \bar{S}_{ij} \frac{\partial w_i^*}{\partial x_j} d^3 \mathbf{x},$$

Non-gravitating case

Potential energy form simplifies such that:

$$\langle \mathbf{u} | H | \mathbf{u} \rangle = \frac{1}{4} \int_M A_{ijkl} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right)^* \left(\frac{\partial u_k}{\partial x_l} + \frac{\partial u_l}{\partial x_k} \right) d^3 \mathbf{x},$$

Non-negative values so long as we have **uniform pointwise stability**:

$$A_{ijkl}(\mathbf{x}) e_{ij}^* e_{kl} \geq k e_{ij}^* e_{ij}$$

Implies elastic wave speeds are all positive

In isotropic case, implies P-wave speed greater than for S-waves

Gravitational contribution

Quantitative treatment possible, but difficult

This shows that there can be a finite number of eigenspaces with negative squared eigenfrequencies

Existence depends on interplay between gravitational and elastic forces

Elastic forces always resist further deformation

Gravitational forces can encourage it

A spherically symmetric compression would lower the (always negative) gravitational binding energy, while raises the elastic potential energy

